

Contents

Executive Summary	1
Lab 4: Intrusion Detection	2
Bibliography	10
Appendix	11

Executive Summary

For this lab we learned about SNORT while primarily utilizing it on a Windows VM. The first section focused on the configuration of SNORT such as identifying the correct network interface and then ensuring it was correct. Once SNORT was set up properly, it was tested against traffic from a Kali machine using ping, netcat, and mnap. This traffic was captured in binary format and analyzed using Wireshark.

Once that was complete, we took a closer look at SNORT's rules and syntax structure. To do so, we picked two CVEs and wrote our own rules for SNORT to use against them. This process helped build a stronger understanding of SNORT and how it could be utilized in a real-world scenario.

Lab 4: Intrusion Detection

In this lab we are investigating SNORT and its uses as an IDS software primarily on a Windows machine. First, we will start the software, then we will examine the configuration options. Next, we will view the traffic in WireShark. Finally, we will write our own custom SNORT rules to try on our own. “Snort is the foremost Open Source Intrusion Prevention System (IPS) in the world. Snort IPS uses a series of rules that help define malicious network activity and uses those rules to find packets that match against them and generates alerts for users” (Snort). Being an open-source software means that it is available to anyone for free, which can be helpful for the every day user or a small business who is unable to pay large fees for an IDS. As long as SNORT is configured properly, it can be a very effective tool in the cybersecurity world. Not only that, but it can be used as the basis to build one’s own IPS software, although that does require a bit more expertise.

To start off, determining the correct network interface SNORT will use is important because SNORT needs to be able to capture the correct traffic in order to be effective. When SNORT is on the wrong interface, it won’t be able to detect the threats it’s supposed to, rendering it pointless. Setting SNORT up on the right interface is the first step to protection and getting valuable information.

Figure 1 shows an image of the network interface configuration. In order to see this, the command is “snort -W” which lists the available interfaces on Windows machines. The Index is the ID number used to communicate with SNORT. The Physical Address section shows the MAC address of the device. The IP address section shows the IP address of the device. The Device Name section shows the name of the device.

Lastly, the description is a human readable device name. When SNORT is run in-line, the network configuration will look the same.

In Figure 2, I tested and validated the configuration file by using the command “snort -T -c C:\Users\Administrator\Desktop\Snort\etc\snort.conf -i 3”. The -T flag tests and reports on the current SNORT configuration. The -c flag specifies the rules file. The -i flag tells SNORT what interface to listen on. I picked interface 3 because that is the machine that I am using with the IP address: 172.16.3.18. Interface 2 in this scenario was the control provided by SimSpace and would not have rendered useful when running SNORT.

Figure 3 shows SNORT having been started and processing packets. In order to start it, I used the command “snort -A console -i 3 -c C:\Users\Administrator\Desktop\Snort\etc\snort.conf -l (L) C:\Users\Administrator\Desktop\Snort\log”. The -A flag tells SNORT what alert mode to run out of several options: fast, full, console, test, or none. For this instance, I just picked console because I wasn’t sure what I needed yet or how SNORT needed to be run for later in the lab. Next, the -l (L) flag tells SNORT to log its findings to a specific directory which can be found once SNORT is finished running.

The configuration files for SNORT are found in the \etc directory under SNORT. There are lots of ways to customize SNORT and make it work the way you want it to. The .conf file shown in Figure 4 has several samples to look at to get a better idea of how these files work. Nine examples are listed which are: set the network variables, configure the decoder, configure the base detection engine, configure dynamic loaded libraries, configure preprocessors, configure output plugins, customize your rule set,

customize preprocessor and decoder rule set, customize shared object rule set.

Setting the network variables defines what network SNORT will be protecting. This is important because without the correct networks identified, SNORT could miss attacks. Configuring the decoder prepares raw data for inspection. This process can detect malicious packets early in the process. Configuring the base detection engine tells SNORT how to match packets to rules. This can affect performance and accuracy. Configuring Dynamic loaded libraries loads the dynamic modules to get them ready for detection. Configuring preprocessors gets them set up to analyze packets before they reach the detection engine. Protocols are analyzed before the rules are applied to detect specific types of attacks. Configuring output plugins tells SNORT where to send the alerts and logs it gets. This promotes organization and can help with integration later on. Customizing the rule set selects which rules SNORT will use. This will tailor detection for a specific environment and make SNORT more accurate. Customizing the preprocessor and decoder rule set are more specific rules about alerts. This helps single out the more severe alerts that could be lost in a lot of lower priority ones. Customizing shared object rule sets gets the advanced rules ready. These shared object rules are powerful detection tools that will help detection.

Figure 5 shows the rules included in the .conf file. Below these rules are the preprocessor and decoder rule set and the shared object rule set. All three of these sections define rules for SNORT. By looking at the rules that are not commented out in the .conf file, I determined that SNORT is configured for IDS mode currently. This is because when you view the rule file, the rules start with “alert”. This indicates IDS

instead of IPS. Rules geared towards IPS would start with words like “block”, “reject”, or “drop”. An example of the app-detect rule is shown in Figure 6.

In order to change the location of the rules, I would add the line “var RULE_PATH filepath”. This tells SNORT where to find the rules within the directory. Then down in the snort.conf file I would write “include \$RULE_PATH/example.rules”. This tells snort the rule to call on when it runs. In order to change SNORT to IPS, the rules would have to be changed from “alert” to “block”, “reject”, or “drop”. Then when running SNORT, it would need to be started in inline mode instead of IDS.

For this portion of the lab we had to run SNORT on the Windows VM while sending traffic to it from another. Then, we had to take the log we generated and put it into WireShark to analyze. Starting off with this section, I had to figure out the best way to run SNORT and I determined that running it without any rules would be fine for this test so I could see all traffic. The command I put into the terminal was “snort -i 3 -b -l C:\Users\Administrator\Desktop\Snort\log”. The -b flag told SNORT to make it a binary file, which would make it usable in WireShark. Once SNORT was running, I used my Kali VM (14 for reference) to send traffic. I started by pinging my machine using “ping 172.16.3.18”. Then I used the command “nc 172.16.3.18 80”. Lastly, I used “nmap -sS 172.16.3.18”. Between these three, it was enough to generate some packets for WireShark, shown in Figure 7.

Starting off, the first several packets (12-34) are the pings that I sent from the Kali machine (IP: 172.16.3.113) to the Windows machine (IP: 172.16.3.18). Pings are not an odd thing to see in a Wireshark capture and could be perfectly normal. Some normal reason to see a ping would be to check round-trip time or to make sure that the host is

available. Though when they aren't normal is when they occur in a high volume, such as an ICMP flood. "An Internet Control Message Protocol (ICMP) flood DDoS attack, also known as a Ping flood attack, is a common Denial-of-Service (DoS) attack in which an attacker attempts to overwhelm a targeted device with ICMP echo-requests (pings)" (NetScout). It is also abnormal to see ICMP echo requests that go out to multiple hosts or sequential IP addresses from one source. This behavior could indicate the start of an attack.

Next, I used netcat and attempted to connect to Port 80 on the Windows machine (packet 38). When I ran this, nothing ever popped up so I can only assume it was unsuccessful – or perhaps I didn't wait long enough. Either way, the only thing that occurred in Wireshark was a TCP packet from the Kali machine to the Windows machine. Typically, there would be a TCP three-way handshake, though the handshake ended up never occurring. "Port 80 is the default network port for web servers using HTTP. It operates on the application layer of the TCP/IP networking model and serves as the communication gateway for HTTP requests and responses between client computers and servers" (NordVPN). Seeing an attempt to connect to Port 80 could be a major red flag and it's important to monitor what happens next as it could indicate an attack, especially depending on where the request came from.

Lastly, I used nmap to attempt a TCP SYN scan which is a half-open scan, shown in Figure 8. "This technique is often referred to as half-open scanning, because you don't open a full TCP connection. You send a SYN packet, as if you are going to open a real connection and then wait for a response. A SYN/ACK indicates the port is listening (open), while a RST (reset) is indicative of a non-listener. If no response is

received after several retransmissions, the port is marked as filtered” (Lyon). From packet 63 all the way to 2078, this scan occurred. This would be a major red flag as it’s a good way for an attacker to tell what ports are open and potentially unprotected. To a defense team, it could be potentially indicative of someone preparing to enter a system. It would be important to see which ports were reported back as open so action could be taken quickly.

Overall, all of these alone could potentially be normal traffic, depending on the situation. However, it is important to note them and look for other indicators of signs of intrusion. All of these together, would certainly be concerning in a real scenario. Wireshark is such an essential tool when investigating happenings like this.

Snort is a very flexible tool that allows for users to write their own rules as needed. “While Snort rules are usually written in a single line, recent versions of Snort allow for multi-line rules; this is especially useful for more sophisticated rules that can be difficult to restrict to just one line. Snort rules consist of two logical parts: a rule header and rule options” (Lenaerts-Bergmans). Lenaerts-Bergmans goes on to say that header consists of an action to take, IP addresses, port numbers, direction of traffic, and inspection protocol. The rule options consist of a message to display when the rule matches, flow state, content or pattern, service or application protocol, and Snort ID and revision number. These components are important to remember when writing custom rules because otherwise, they likely will not work.

A CVE stands for common vulnerabilities and exposures. To pick two CVEs, I went to cisa.gov and reviewed the list that they keep. For my first CVE I picked CVE-2025-24054 named the Microsoft Windows NTLM Hash Disclosure Spoofing

Vulnerability. This vulnerability is described as “Microsoft Windows NTLM contains an external control of file name or path vulnerability that allows an unauthorized attacker to perform spoofing over a network” (CISA). I am not extremely knowledgeable on SNORT rules and did need a little assistance from ChatGPT. The rule I wrote was: “alert tcp \$HOME_NET any -> !\$HOME_NET 445 (msg:“Possible CVE-2025-24054 NTLM hash leak via SMB”; flow:established,to_server; content:“NTLMSSP”; offset:0; depth:100; sid:1000001; rev:1;)”. This rule alerts on any outbound TCP traffic on port 445, which is used for SMB. The flow applies to established connections directed to a server. The content portion is looking for NTLM authentication attempts. The offset and depth is looking at the first 100 bytes of the packet where the signature for NTLMSSP would be. Lastly, a SNORT ID and revision ID is assigned. This rule would not definitively identify this CVE but it would be a step in the right direction and more investigation would need to be done to confirm. Figure 9 shows the rules file.

My next CVE I chose is CVE-2025-31161 named CrushFTP Authentication Bypass Vulnerability. This is defined as “CrushFTP contains an authentication bypass vulnerability in the HTTP authorization header that allows a remote unauthenticated attacker to authenticate to any known or guessable user account (e.g., crushadmin), potentially leading to a full compromise” (CISA) The rule I wrote was “alert tcp \$EXTERNAL_NET any -> \$HOME_NET [80,443] (msg:“CVE-2025-31161 CrushFTP Authentication Bypass Attempt”; flow:to_server,established; content:“Authorization: AWS4-HMAC-SHA256 Credential=”; nocase; http_header; content:“/”; within:50; http_header; content:“Cookie: CrushAuth=”; nocase; http_header; classtype:attempted-admin; sid:1000003; rev:1;)”. This rule is an alert that alerts for any TCP traffic coming

from an external source on port 80 and 443 which are for HTTP and HTTPS, respectively. The flow is the same as above which is established connections. The content detects the specific authentication header used and the http header portion states that the content portion should be found in the header. The forward slash should follow the "Credential=" within 50 bytes. The next content portion is checking for the CrushAuth cookie which is also part of the exploit. The class type is an attempted administrative access. This rule covers most of the bases but could potentially cause false positives or may miss some attempts in this vulnerability. The rule is shown in Figure 10 and Figure 11 shows SNORT running successfully with both rules in place.

This lab was a good opportunity to learn about some basics involving intrusion detection. Not only that but it provided information over the tool SNORT which has lots of uses and can assist when intrusions occur in real time. Overall, SNORT itself was relatively easy to understand but the rules themselves were not the easiest things to write.

Bibliography

CISA. "Known Exploited Vulnerabilities Catalog | CISA." *Www.cisa.gov*,

www.cisa.gov/known-exploited-vulnerabilities-catalog?page=1.

Lenaerts-Bergmans, Bart. "Snort Explained: Understanding Snort Rules and Use Cases

| CrowdStrike." *CrowdStrike.com*, 2024, [www.crowdstrike.com/en-](https://www.crowdstrike.com/en-us/cybersecurity-101/threat-intelligence/snort-rules/)

[us/cybersecurity-101/threat-intelligence/snort-rules/](https://www.crowdstrike.com/en-us/cybersecurity-101/threat-intelligence/snort-rules/).

Lyon, Gordon. "Port Scanning Techniques | Nmap Network Scanning." *Nmap.org*, 2019,

nmap.org/book/man-port-scanning-techniques.html.

NetScout. "What Is an ICMP Flood Attack?" *NETSCOUT*, 2025,

www.netscout.com/what-is-ddos/icmp-flood?utm_source=chatgpt.com. Accessed

14 May 2025.

NordVPN. "Port 80 Definition – Glossary | NordVPN." *Nordvpn.com*, 22 Aug. 2023,

nordvpn.com/cybersecurity/glossary/port-80/.

Snort. "Snort - Network Intrusion Detection & Prevention System." *Snort.org*, Snort,

2019, www.snort.org/.

Appendix

```

C:\Users\Administrator\Desktop\Snort\bin>snort -W

-*> Snort! <*-
o"~
....~
Version 2.9.17-WIN32 GRE (Build 199)
By Martin Roesch & The Snort Team: http://www.snort.org/contact#team
Copyright (C) 2014-2020 Cisco and/or its affiliates. All rights reserved.
Copyright (C) 1998-2013 Sourcefire, Inc., et al.
Using PCRE version: 8.10 2010-06-25
Using ZLIB version: 1.2.3

Index  Physical Address      IP Address      Device Name      Description
-----
1      00:00:00:00:00:00        0.0.0.0 \Device\NPF_{1DB0560C-8B99-4D04-8E95-B8A670920723} MS NDIS 6.0 LoopBack Driver
2      00:00:00:00:00:00        10.10.0.23 \Device\NPF_{9A7A2AB6-BB6C-40B2-B74F-377F03513C60} VMware vmxnet3 virtual network device
3      00:00:00:00:00:00        172.16.3.18 \Device\NPF_{B09EB7B3-148C-446C-B745-B2BB82CED37A} VMware vmxnet3 virtual network device

C:\Users\Administrator\Desktop\Snort\bin>

```

Figure 1: Network interface configuration

```

+-----+
[ Number of patterns truncated to 20 bytes: 452 ]
pcap DAQ configured to passive.
The DAQ version does not support reload.
Acquiring network traffic from "\Device\NPF_{B09EB7B3-148C-446C-B745-B2BB82CED37A}".

---- Initialization Complete ----

-*> Snort! <*-
o"~
....~
Version 2.9.17-WIN32 GRE (Build 199)
By Martin Roesch & The Snort Team: http://www.snort.org/contact#team
Copyright (C) 2014-2020 Cisco and/or its affiliates. All rights reserved.
Copyright (C) 1998-2013 Sourcefire, Inc., et al.
Using PCRE version: 8.10 2010-06-25
Using ZLIB version: 1.2.3

Rules Engine: SF_SNORT_DETECTION_ENGINE Version 3.1 <Build 1>
Preprocessor Object: SF_SSLPP Version 1.1 <Build 4>
Preprocessor Object: SF_SSH Version 1.1 <Build 3>
Preprocessor Object: SF_SMTP Version 1.1 <Build 9>
Preprocessor Object: SF_SIP Version 1.1 <Build 1>
Preprocessor Object: SF_SDF Version 1.1 <Build 1>
Preprocessor Object: SF_REPUTATION Version 1.1 <Build 1>
Preprocessor Object: SF_POP Version 1.0 <Build 1>
Preprocessor Object: SF_MODBUS Version 1.1 <Build 1>
Preprocessor Object: SF_IMAP Version 1.0 <Build 1>
Preprocessor Object: SF_GTP Version 1.1 <Build 1>
Preprocessor Object: SF_FTPTELNET Version 1.2 <Build 13>
Preprocessor Object: SF_DNS Version 1.1 <Build 4>
Preprocessor Object: SF_DNP3 Version 1.1 <Build 1>
Preprocessor Object: SF_DCERPC2 Version 1.0 <Build 3>

Snort successfully validated the configuration!
Snort exiting

```

Figure 2: Tested and validated configuration file

```

[ Number of patterns truncated to 20 bytes: 452 ]
pcap DAQ configured to passive.
The DAQ version does not support reload.
Acquiring network traffic from "\Device\NPF_{B09EB7B3-148C-446C-B745-B2BB82CED37A}".
Decoding Ethernet

--- Initialization Complete ---

o'-'~
'''~
-*) Snort! <*-
Version 2.9.17-WIN32 GRE (Build 199)
By Martin Roesch & The Snort Team: http://www.snort.org/contact#team
Copyright (C) 2014-2020 Cisco and/or its affiliates. All rights reserved.
Copyright (C) 1998-2013 Sourcefire, Inc., et al.
Using PCRE version: 8.10 2010-06-25
Using ZLIB version: 1.2.3

Rules Engine: SF_SNORT_DETECTION_ENGINE Version 3.1 <Build 1>
Preprocessor Object: SF_SSLPP Version 1.1 <Build 4>
Preprocessor Object: SF_SSH Version 1.1 <Build 3>
Preprocessor Object: SF_SMTP Version 1.1 <Build 9>
Preprocessor Object: SF_SIP Version 1.1 <Build 1>
Preprocessor Object: SF_SDF Version 1.1 <Build 1>
Preprocessor Object: SF_REPUTATION Version 1.1 <Build 1>
Preprocessor Object: SF_POP Version 1.0 <Build 1>
Preprocessor Object: SF_MODBUS Version 1.1 <Build 1>
Preprocessor Object: SF_IMAP Version 1.0 <Build 1>
Preprocessor Object: SF_GTP Version 1.1 <Build 1>
Preprocessor Object: SF_FTPTELNET Version 1.2 <Build 13>
Preprocessor Object: SF_DNS Version 1.1 <Build 4>
Preprocessor Object: SF_DNP3 Version 1.1 <Build 1>
Preprocessor Object: SF_DCERPC2 Version 1.0 <Build 3>
Commencing packet processing (pid=4128)

```

Figure 3: SNORT started

```

# The .conf format file Map
#
# This configuration file enables active response, to run snort in
# test mode. If you are required to supply an interface -i interface
# or test mode will fail to fully validate the configuration and
# exit with a fatal error.
#-----
#
# This file contains a sample snort configuration.
# You should take the following steps to create your own custom configuration:
#
# 1) Set the network variables.
# 2) Configure the decoder.
# 3) Configure the base detection engine.
# 4) Configure dynamic loaded libraries.
# 5) Configure preprocessors.
# 6) Configure output plugins.
# 7) Customize your rule set.
# 8) Customize preprocessor and decoder rule set.
# 9) Customize shared object rule set.
#-----
#
# Step #1: Set the network variables. For more information, see README.variables
#-----
#
# Setup the network addresses you are protecting
ipvar HOME_NET 172.16.3.0/24
#
# Set up the external network addresses. Leave as "any" in most situations
ipvar EXTERNAL_NET any
#
# List of DNS servers on your network
ipvar DNS_SERVERS $HOME_NET
#
# List of SMTP servers on your network
ipvar SMTP_SERVERS $HOME_NET
#
# List of web servers on your network
ipvar WWW_SERVERS $HOME_NET
#
# List of sql servers on your network
ipvar SQL_SERVERS $HOME_NET
#
# List of telnet servers on your network
ipvar TELNET_SERVERS $HOME_NET
#
# List of ssh servers on your network
ipvar SSH_SERVERS $HOME_NET
#
# List of ftp servers on your network
ipvar FTP_SERVERS $HOME_NET
#
# List of ports you run web servers on
portvar HTTP_PORTS {80,81,301,302,303,304,305,306,307,308,309,310,311,312,313,314,315,316,317,318,319,320,321,322,323,324,325,326,327,328,329,330,331,332,333,334,335,336,337,338,339,340,341,342,343,344,345,346,347,348,349,350,351,352,353,354,355,356,357,358,359,360,361,362,363,364,365,366,367,368,369,370,371,372,373,374,375,376,377,378,379,380,381,382,383,384,385,386,387,388,389,390,391,392,393,394,395,396,397,398,399,400,401,402,403,404,405,406,407,408,409,410,411,412,413,414,415,416,417,418,419,420,421,422,423,424,425,426,427,428,429,430,431,432,433,434,435,436,437,438,439,440,441,442,443,444,445,446,447,448,449,450,451,452,453,454,455,456,457,458,459,460,461,462,463,464,465,466,467,468,469,470,471,472,473,474,475,476,477,478,479,480,481,482,483,484,485,486,487,488,489,490,491,492,493,494,495,496,497,498,499,500,501,502,503,504,505,506,507,508,509,510,511,512,513,514,515,516,517,518,519,520,521,522,523,524,525,526,527,528,529,530,531,532,533,534,535,536,537,538,539,540,541,542,543,544,545,546,547,548,549,550,551,552,553,554,555,556,557,558,559,560,561,562,563,564,565,566,567,568,569,570,571,572,573,574,575,576,577,578,579,580,581,582,583,584,585,586,587,588,589,590,591,592,593,594,595,596,597,598,599,600,601,602,603,604,605,606,607,608,609,610,611,612,613,614,615,616,617,618,619,620,621,622,623,624,625,626,627,628,629,630,631,632,633,634,635,636,637,638,639,640,641,642,643,644,645,646,647,648,649,650,651,652,653,654,655,656,657,658,659,660,661,662,663,664,665,666,667,668,669,670,671,672,673,674,675,676,677,678,679,680,681,682,683,684,685,686,687,688,689,690,691,692,693,694,695,696,697,698,699,700,701,702,703,704,705,706,707,708,709,710,711,712,713,714,715,716,717,718,719,720,721,722,723,724,725,726,727,728,729,730,731,732,733,734,735,736,737,738,739,740,741,742,743,744,745,746,747,748,749,750,751,752,753,754,755,756,757,758,759,760,761,762,763,764,765,766,767,768,769,770,771,772,773,774,775,776,777,778,779,780,781,782,783,784,785,786,787,788,789,790,791,792,793,794,795,796,797,798,799,800,801,802,803,804,805,806,807,808,809,810,811,812,813,814,815,816,817,818,819,820,821,822,823,824,825,826,827,828,829,830,831,832,833,834,835,836,837,838,839,840,841,842,843,844,845,846,847,848,849,850,851,852,853,854,855,856,857,858,859,860,861,862,863,864,865,866,867,868,869,870,871,872,873,874,875,876,877,878,879,880,881,882,883,884,885,886,887,888,889,890,891,892,893,894,895,896,897,898,899,900,901,902,903,904,905,906,907,908,909,910,911,912,913,914,915,916,917,918,919,920,921,922,923,924,925,926,927,928,929,930,931,932,933,934,935,936,937,938,939,940,941,942,943,944,945,946,947,948,949,950,951,952,953,954,955,956,957,958,959,960,961,962,963,964,965,966,967,968,969,970,971,972,973,974,975,976,977,978,979,980,981,982,983,984,985,986,987,988,989,990,991,992,993,994,995,996,997,998,999,1000,1001,1002,1003,1004,1005,1006,1007,1008,1009,1010,1011,1012,1013,1014,1015,1016,1017,1018,1019,1020,1021,1022,1023,1024,1025,1026,1027,1028,1029,1030,1031,1032,1033,1034,1035,1036,1037,1038,1039,1040,1041,1042,1043,1044,1045,1046,1047,1048,1049,1050,1051,1052,1053,1054,1055,1056,1057,1058,1059,1060,1061,1062,1063,1064,1065,1066,1067,1068,1069,1070,1071,1072,1073,1074,1075,1076,1077,1078,1079,1080,1081,1082,1083,1084,1085,1086,1087,1088,1089,1090,1091,1092,1093,1094,1095,1096,1097,1098,1099,1100,1101,1102,1103,1104,1105,1106,1107,1108,1109,1110,1111,1112,1113,1114,1115,1116,1117,1118,1119,1120,1121,1122,1123,1124,1125,1126,1127,1128,1129,1130,1131,1132,1133,1134,1135,1136,1137,1138,1139,1140,1141,1142,1143,1144,1145,1146,1147,1148,1149,1150,1151,1152,1153,1154,1155,1156,1157,1158,1159,1160,1161,1162,1163,1164,1165,1166,1167,1168,1169,1170,1171,1172,1173,1174,1175,1176,1177,1178,1179,1180,1181,1182,1183,1184,1185,1186,1187,1188,1189,1190,1191,1192,1193,1194,1195,1196,1197,1198,1199,1200,1201,1202,1203,1204,1205,1206,1207,1208,1209,1210,1211,1212,1213,1214,1215,1216,1217,1218,1219,1220,1221,1222,1223,1224,1225,1226,1227,1228,1229,1230,1231,1232,1233,1234,1235,1236,1237,1238,1239,1240,1241,1242,1243,1244,1245,1246,1247,1248,1249,1250,1251,1252,1253,1254,1255,1256,1257,1258,1259,1260,1261,1262,1263,1264,1265,1266,1267,1268,1269,1270,1271,1272,1273,1274,1275,1276,1277,1278,1279,1280,1281,1282,1283,1284,1285,1286,1287,1288,1289,1290,1291,1292,1293,1294,1295,1296,1297,1298,1299,1300,1301,1302,1303,1304,1305,1306,1307,1308,1309,1310,1311,1312,1313,1314,1315,1316,1317,1318,1319,1320,1321,1322,1323,1324,1325,1326,1327,1328,1329,1330,1331,1332,1333,1334,1335,1336,1337,1338,1339,1340,1341,1342,1343,1344,1345,1346,1347,1348,1349,1350,1351,1352,1353,1354,1355,1356,1357,1358,1359,1360,1361,1362,1363,1364,1365,1366,1367,1368,1369,1370,1371,1372,1373,1374,1375,1376,1377,1378,1379,1380,1381,1382,1383,1384,1385,1386,1387,1388,1389,1390,1391,1392,1393,1394,1395,1396,1397,1398,1399,1400,1401,1402,1403,1404,1405,1406,1407,1408,1409,1410,1411,1412,1413,1414,1415,1416,1417,1418,1419,1420,1421,1422,1423,1424,1425,1426,1427,1428,1429,1430,1431,1432,1433,1434,1435,1436,1437,1438,1439,1440,1441,1442,1443,1444,1445,1446,1447,1448,1449,1450,1451,1452,1453,1454,1455,1456,1457,1458,1459,1460,1461,1462,1463,1464,1465,1466,1467,1468,1469,1470,1471,1472,1473,1474,1475,1476,1477,1478,1479,1480,1481,1482,1483,1484,1485,1486,1487,1488,1489,1490,1491,1492,1493,1494,1495,1496,1497,1498,1499,1500,1501,1502,1503,1504,1505,1506,1507,1508,1509,1510,1511,1512,1513,1514,1515,1516,1517,1518,1519,1520,1521,1522,1523,1524,1525,1526,1527,1528,1529,1530,1531,1532,1533,1534,1535,1536,1537,1538,1539,1540,1541,1542,1543,1544,1545,1546,1547,1548,1549,1550,1551,1552,1553,1554,1555,1556,1557,1558,1559,1560,1561,1562,1563,1564,1565,1566,1567,1568,1569,1570,1571,1572,1573,1574,1575,1576,1577,1578,1579,1580,1581,1582,1583,1584,1585,1586,1587,1588,1589,1590,1591,1592,1593,1594,1595,1596,1597,1598,1599,1600,1601,1602,1603,1604,1605,1606,1607,1608,1609,1610,1611,1612,1613,1614,1615,1616,1617,1618,1619,1620,1621,1622,1623,1624,1625,1626,1627,1628,1629,1630,1631,1632,1633,1634,1635,1636,1637,1638,1639,1640,1641,1642,1643,1644,1645,1646,1647,1648,1649,1650,1651,1652,1653,1654,1655,1656,1657,1658,1659,1660,1661,1662,1663,1664,1665,1666,1667,1668,1669,1670,1671,1672,1673,1674,1675,1676,1677,1678,1679,1680,1681,1682,1683,1684,1685,1686,1687,1688,1689,1690,1691,1692,1693,1694,1695,1696,1697,1698,1699,1700,1701,1702,1703,1704,1705,1706,1707,1708,1709,1710,1711,1712,1713,1714,1715,1716,1717,1718,1719,1720,1721,1722,1723,1724,1725,1726,1727,1728,1729,1730,1731,1732,1733,1734,1735,1736,1737,1738,1739,1740,1741,1742,1743,1744,1745,1746,1747,1748,1749,1750,1751,1752,1753,1754,1755,1756,1757,1758,1759,1760,1761,1762,1763,1764,1765,1766,1767,1768,1769,1770,1771,1772,1773,1774,1775,1776,1777,1778,1779,1780,1781,1782,1783,1784,1785,1786,1787,1788,1789,1790,1791,1792,1793,1794,1795,1796,1797,1798,1799,1800,1801,1802,1803,1804,1805,1806,1807,1808,1809,1810,1811,1812,1813,1814,1815,1816,1817,1818,1819,1820,1821,1822,1823,1824,1825,1826,1827,1828,1829,1830,1831,1832,1833,1834,1835,1836,1837,1838,1839,1840,1841,1842,1843,1844,1845,1846,1847,1848,1849,1850,1851,1852,1853,1854,1855,1856,1857,1858,1859,1860,1861,1862,1863,1864,1865,1866,1867,1868,1869,1870,1871,1872,1873,1874,1875,1876,1877,1878,1879,1880,1881,1882,1883,1884,1885,1886,1887,1888,1889,1890,1891,1892,1893,1894,1895,1896,1897,1898,1899,1900,1901,1902,1903,1904,1905,1906,1907,1908,1909,1910,1911,1912,1913,1914,1915,1916,1917,1918,1919,1920,1921,1922,1923,1924,1925,1926,1927,1928,1929,1930,1931,1932,1933,1934,1935,1936,1937,1938,1939,1940,1941,1942,1943,1944,1945,1946,1947,1948,1949,1950,1951,1952,1953,1954,1955,1956,1957,1958,1959,1960,1961,1962,1963,1964,1965,1966,1967,1968,1969,1970,1971,1972,1973,1974,1975,1976,1977,1978,1979,1980,1981,1982,1983,1984,1985,1986,1987,1988,1989,1990,1991,1992,1993,1994,1995,1996,1997,1998,1999,2000,2001,2002,2003,2004,2005,2006,2007,2008,2009,2010,2011,2012,2013,2014,2015,2016,2017,2018,2019,2020,2021,2022,2023,2024,2025,2026,2027,2028,2029,2030,2031,2032,2033,2034,2035,2036,2037,2038,2039,2040,2041,2042,2043,2044,2045,2046,2047,2048,2049,2050,2051,2052,2053,2054,2055,2056,2057,2058,2059,2060,2061,2062,2063,2064,2065,2066,2067,2068,2069,2070,2071,2072,2073,2074,2075,2076,2077,2078,2079,2080,2081,2082,2083,2084,2085,2086,2087,2088,2089,2090,2091,2092,2093,2094,2095,2096,2097,2098,2099,2100,2101,2102,2103,2104,2105,2106,2107,2108,2109,2110,2111,2112,2113,2114,2115,2116,2117,2118,2119,2120,2121,2122,2123,2124,2125,2126,2127,2128,2129,2130,2131,2132,2133,2134,2135,2136,2137,2138,2139,2140,2141,2142,2143,2144,2145,2146,2147,2148,2149,2150,2151,2152,2153,2154,2155,2156,2157,2158,2159,2160,2161,2162,2163,2164,2165,2166,2167,2168,2169,2170,2171,2172,2173,2174,2175,2176,2177,2178,2179,2180,2181,2182,2183,2184,2185,2186,2187,2188,2189,2190,2191,2192,2193,2194,2195,2196,2197,2198,2199,2200,2201,2202,2203,2204,2205,2206,2207,2208,2209,2210,2211,2212,2213,2214,2215,2216,2217,2218,2219,2220,2221,2222,2223,2224,2225,2226,2227,2228,2229,2230,2231,2232,2233,2234,2235,2236,2237,2238,2239,2240,2241,2242,2243,2244,2245,2246,2247,2248,2249,2250,2251,2252,2253,2254,2255,2256,2257,2258,2259,2260,2261,2262,2263,2264,2265,2266,2267,2268,2269,2270,2271,2272,2273,2274,2275,2276,2277,2278,2279,2280,2281,2282,2283,2284,2285,2286,2287,2288,2289,2290,2291,2292,2293,2294,2295,2296,2297,2298,2299,2300,2301,2302,2303,2304,2305,2306,2307,2308,2309,2310,2311,2312,2313,2314,2315,2316,2317,2318,2319,2320,2321,2322,2323,2324,2325,2326,2327,2328,2329,2330,2331,2332,2333,2334,2335,2336,2337,2338,2339,2340,2341,2342,2343,2344,2345,2346,2347,2348,2349,2350,2351,2352,2353,2354,2355,2356,2357,2358,2359,2360,2361,2362,2363,2364,2365,2366,2367,2368,2369,2370,2371,2372,2373,2374,2375,2376,2377,2378,2379,2380,2381,2382,2383,2384,2385,2386,2387,2388,2389,2390,2391,2392,2393,2394,2395,2396,2397,2398,2399,2400,2401,2402,2403,2404,2405,2406,2407,2408,2409,2410,2411,2412,2413,2414,2415,2416,2417,2418,2419,2420,2421,2422,2423,2424,2425,2426,2427,2428,2429,2430,2431,2432,2433,2434,2435,2436,2437,2438,2439,2440,2441,2442,2443,2444,2445,2446,2447,2448,2449,2450,2451,2452,2453,2454,2455,2456,2457,2458,2459,2460,2461,2462,2463,2464,2465,2466,2467,2468,2469,2470,2471,2472,2473,2474,2475,2476,2477,2478,2479,2480,2481,2482,2483,2484,2485,2486,2487,2488,2489,2490,2491,2492,2493,2494,2495,2496,2497,2498,2499,2500,2501,2502,2503,2504,2505,2506,2507,2508,2509,2510,2511,2512,2513,2514,2515,2516,2517,2518,2519,2520,2521,2522,2523,2524,2525,2526,2527,2528,2529,2530,2531,2532,2533,2534,2535,2536,2537,2538,2539,2540,2541,2542,2543,2544,2545,2546,2547,2548,2549,2550,2551,2552,2553,2554,2555,2556,2557,2558,2559,2560,2561,2562,2563,2564,2565,2566,2567,2568,2569,2570,2571,2572,2573,2574,2575,2576,2577,2578,2579,2580,2581,2582,2583,2584,2585,2586,2587,2588,2589,2590,2591,2592,2593,2594,2595,2596,2597,2598,2599,2600,2601,2602,2603,2604,2605,2606,2607,2608,2609,2610,2611,2612,2613,2614,2615,2616,2617,2618,2619,2620,2621,2622,2623,2624,2625,2626,2627,2628,2629,2630,2631,2632,2633,2634,2635,2636,2637,2638,2639,2640,2641,2642,2643,2644,2645,2646,2647,2648,2649,2650,2651,2652,2653,2654,2655,2656,2657,2658,2659,2660,2661,2662,2663,2664,2665,2666,2667,2668,2669,2670,2671,2672,2673,2674,2675,2676,2677,2678,2679,2680,2681,2682,2683,2684,2685,2686,2687,2688,268
```

```
#####
# Step #7: Customize your rule set
# For more information, see Snort Manual, Writing !
#
# NOTE: All categories are enabled in this conf fi
#####

# site specific rules
include $RULE_PATH\local.rules

include $RULE_PATH\app-detect.rules
include $RULE_PATH\attack-responses.rules
include $RULE_PATH\backdoor.rules
include $RULE_PATH\bad-traffic.rules
include $RULE_PATH\blacklist.rules
include $RULE_PATH\botnet-cnc.rules
include $RULE_PATH\browser-chrome.rules
include $RULE_PATH\browser-firefox.rules
include $RULE_PATH\browser-ie.rules
include $RULE_PATH\browser-other.rules
include $RULE_PATH\browser-plugins.rules
include $RULE_PATH\browser-webkit.rules
include $RULE_PATH\chat.rules
include $RULE_PATH\content-replace.rules
include $RULE_PATH\ddos.rules
include $RULE_PATH\dns.rules
include $RULE_PATH\dos.rules
include $RULE_PATH\experimental.rules
include $RULE_PATH\exploit-kit.rules
include $RULE_PATH\exploit.rules
include $RULE_PATH\file-executable.rules
include $RULE_PATH\file-flash.rules
include $RULE_PATH\file-identify.rules
include $RULE_PATH\file-image.rules
include $RULE_PATH\file-multimedia.rules
include $RULE_PATH\file-office.rules
include $RULE_PATH\file-other.rules
include $RULE_PATH\file-pdf.rules
include $RULE_PATH\finger.rules
include $RULE_PATH\ftp.rules
include $RULE_PATH\icmp-info.rules
include $RULE_PATH\icmp.rules
include $RULE_PATH\imap.rules
include $RULE_PATH\indicator-compromise.rules
include $RULE_PATH\indicator-obfuscation.rules
include $RULE_PATH\indicator-shellcode.rules
include $RULE_PATH\info.rules
include $RULE_PATH\malware-backdoor.rules
include $RULE_PATH\malware-cnc.rules
include $RULE_PATH\malware-other.rules
include $RULE_PATH\malware-tools.rules
include $RULE_PATH\misc.rules
include $RULE_PATH\multimedia.rules
include $RULE_PATH\mysql.rules
#include $RULE_PATH\netbios.rules
include $RULE_PATH\nntp.rules
include $RULE_PATH\oracle.rules
#include $RULE_PATH\os-linux.rules
#include $RULE_PATH\os-other.rules
include $RULE_PATH\os-solaris.rules
include $RULE_PATH\os-windows.rules
include $RULE_PATH\other-ids.rules
include $RULE_PATH\p2p.rules
include $RULE_PATH\phishing-spam.rules
include $RULE_PATH\policy-multimedia.rules
include $RULE_PATH\policy-other.rules
include $RULE_PATH\policy.rules
include $RULE_PATH\policy-social.rules
include $RULE_PATH\policy-spam.rules
include $RULE_PATH\pop2.rules
include $RULE_PATH\pop3.rules
include $RULE_PATH\protocol-finger.rules
include $RULE_PATH\protocol-ftp.rules
#include $RULE_PATH\protocol-icmp.rules
#include $RULE_PATH\protocol-imap.rules
#include $RULE_PATH\protocol-pop.rules
include $RULE_PATH\protocol-services.rules
#include $RULE_PATH\protocol-voip.rules
#include $RULE_PATH\pua-adware.rules
#include $RULE_PATH\pua-other.rules
#include $RULE_PATH\pua-p2p.rules
include $RULE_PATH\pua-toolbars.rules
include $RULE_PATH\rpc.rules
include $RULE_PATH\rservices.rules
include $RULE_PATH\scada.rules
include $RULE_PATH\scan.rules
include $RULE_PATH\server-apache.rules
include $RULE_PATH\server-iis.rules
#include $RULE_PATH\server-mail.rules
#include $RULE_PATH\server-mssql.rules
#include $RULE_PATH\server-mysql.rules
#include $RULE_PATH\server-oracle.rules
#include $RULE_PATH\server-other.rules
include $RULE_PATH\server-webapp.rules
include $RULE_PATH\shellcode.rules
include $RULE_PATH\smtp.rules
include $RULE_PATH\snmp.rules
include $RULE_PATH\specific-threats.rules
include $RULE_PATH\spyware-put.rules
include $RULE_PATH\sql.rules
include $RULE_PATH\telnet.rules
include $RULE_PATH\tftp.rules
include $RULE_PATH\virus.rules
include $RULE_PATH\voip.rules
include $RULE_PATH\web-activex.rules
include $RULE_PATH\web-attacks.rules
include $RULE_PATH\web-cgi.rules
include $RULE_PATH\web-client.rules
include $RULE_PATH\web-coldfusion.rules
include $RULE_PATH\web-frontpage.rules
include $RULE_PATH\web-iis.rules
include $RULE_PATH\web-misc.rules
include $RULE_PATH\web-php.rules
include $RULE_PATH\x11.rules
```

Figure 5: Rules in SNORT.conf file


```

app-detect - Notepad
File Edit Format View Help
# Copyright 2001-2021 Sourcefire, Inc. All Rights Reserved.
#
# This file contains (i) proprietary rules that were created, tested and certified by
# Sourcefire, Inc. (the "VRT Certified Rules") that are distributed under the VRT
# Certified Rules License Agreement (v 2.0), and (ii) rules that were created by
# Sourcefire and other third parties (the "GPL Rules") that are distributed under the
# GNU General Public License (GPL), v2.
#
# The VRT Certified Rules are owned by Sourcefire, Inc. The GPL Rules were created
# by Sourcefire and other third parties. The GPL Rules created by Sourcefire are
# owned by Sourcefire, Inc., and the GPL Rules not created by Sourcefire are owned by
# their respective creators. Please see http://www.snort.org/snort/snort-team/ for a
# list of third party owners and their respective copyrights.
#
# In order to determine what rules are VRT Certified Rules or GPL Rules, please refer
# to the VRT Certified Rules License Agreement (v2.0).
#
#-----
# APP-DETECT RULES
#-----

# alert tcp $HOME_NET any -> $EXTERNAL_NET [80,443,8200] (msg:"APP-DETECT GoToMyPC remote control att
# alert tcp $EXTERNAL_NET any -> $HOME_NET 3389 (msg:"APP-DETECT remote desktop protocol attempted ac
# alert udp any any -> 255.255.255.255 23945 (msg:"APP-DETECT Data Rescue IDA Pro startup license che
alert tcp $HOME_NET [2601:2606] -> $EXTERNAL_NET any (msg:"APP-DETECT Quagga password challenge detec
# alert tcp $HOME_NET any -> $EXTERNAL_NET $HTTP_PORTS (msg:"APP-DETECT HTTP Tunnel proxy outbound cor
# alert udp any 68 -> any 67 (msg:"APP-DETECT Intel AMT DHCP boot request detected"; flow:to_server;
# alert tcp $EXTERNAL_NET any -> $HTTP_SERVERS $HTTP_PORTS (msg:"APP-DETECT OpenVAS Scanner User-Ager
# alert tcp $HOME_NET any -> $EXTERNAL_NET $HTTP_PORTS (msg:"APP-DETECT Bloomberg web crawler outbour
# alert tcp $EXTERNAL_NET any -> $HOME_NET $HTTP_PORTS (msg:"APP-DETECT Jenkins Groovy script access
# alert tcp $HOME_NET any -> $EXTERNAL_NET $HTTP_PORTS (msg:"APP-DETECT Hola VPN startup attempt"; fl
# alert tcp $HOME_NET any -> $EXTERNAL_NET [80,443,8200] (msg:"APP-DETECT Hola VPN tunnel

```

Figure 6: app-detect rule

0.000000	172.16.3.18	172.16.2.3	DNS	86 Standard query 0xc0cd A slscr.update.microsoft.com
2.2.843267	172.16.2.3	172.16.3.18	DNS	86 Standard query response 0xb6fc Server failure A slscr.update.microsoft.com
3.4.016262	172.16.3.18	172.16.2.3	DNS	78 Standard query 0x5ac8 A ISCS-FILE.site.lan
4.4.017282	172.16.2.3	172.16.3.18	DNS	94 Standard query response 0x5ac8 A ISCS-FILE.site.lan A 172.16.2.2
5.4.021633	172.16.3.18	172.16.2.3	DNS	86 Standard query 0x5e8f A slscr.update.microsoft.com
6.4.046898	172.16.3.18	172.16.2.3	DNS	86 Standard query 0x5e8f A slscr.update.microsoft.com
7.5.062404	172.16.3.18	172.16.2.3	DNS	86 Standard query 0x5e8f A slscr.update.microsoft.com
8.7.062553	172.16.3.18	172.16.2.3	DNS	86 Standard query 0x5e8f A slscr.update.microsoft.com
9.7.382131	172.16.2.3	172.16.3.18	DNS	86 Standard query response 0xc0cd Server failure A slscr.update.microsoft.com
10.7.382133	172.16.2.3	172.16.3.18	DNS	86 Standard query response 0xc0cd Server failure A slscr.update.microsoft.com
11.7.382259	172.16.2.3	172.16.3.18	DNS	86 Standard query response 0xc0cd Server failure A slscr.update.microsoft.com
12.7.925964	172.16.3.113	172.16.3.18	ICMP	98 Echo (ping) request id=0x1297, seq=1/256, ttl=64 (reply in 13)
13.7.926136	172.16.3.18	172.16.3.113	ICMP	98 Echo (ping) reply id=0x1297, seq=1/256, ttl=128 (request in 12)
14.8.931181	172.16.3.113	172.16.3.18	ICMP	98 Echo (ping) request id=0x1297, seq=2/512, ttl=64 (reply in 15)
15.8.931247	172.16.3.18	172.16.3.113	ICMP	98 Echo (ping) reply id=0x1297, seq=2/512, ttl=128 (request in 14)
16.9.955891	172.16.3.113	172.16.3.18	ICMP	98 Echo (ping) request id=0x1297, seq=3/768, ttl=64 (reply in 17)
17.9.955250	172.16.3.18	172.16.3.113	ICMP	98 Echo (ping) reply id=0x1297, seq=3/768, ttl=128 (request in 16)
18.10.979185	172.16.3.113	172.16.3.18	ICMP	98 Echo (ping) request id=0x1297, seq=4/1824, ttl=64 (reply in 19)
19.10.979337	172.16.3.18	172.16.3.113	ICMP	98 Echo (ping) reply id=0x1297, seq=4/1824, ttl=128 (request in 18)
20.11.007050	172.16.2.3	172.16.3.18	DNS	86 Standard query response 0xc0cd Server failure A slscr.update.microsoft.com
21.11.078876	172.16.3.18	172.16.2.3	DNS	86 Standard query 0x5e8f A slscr.update.microsoft.com
22.12.005237	172.16.3.113	172.16.3.18	ICMP	98 Echo (ping) request id=0x1297, seq=5/1280, ttl=64 (reply in 23)
23.12.003378	172.16.3.18	172.16.3.113	ICMP	98 Echo (ping) reply id=0x1297, seq=5/1280, ttl=128 (request in 22)
24.12.765420	Intel_00:00:1b Intel_00:07:1b	ARP	42 Who has 172.16.3.113? Tell 172.16.3.18	
25.12.765738	Intel_00:07:1b Intel_00:08:1b	ARP	60 172.16.3.113 is at 00:02:b3:00:07:1b	
26.12.995156	Intel_00:07:1b Intel_00:08:1b	ARP	60 Who has 172.16.3.18? Tell 172.16.3.113	
27.12.995203	Intel_00:08:1b Intel_00:07:1b	ARP	42 172.16.3.18 is at 00:02:b3:00:08:1b	
28.13.027446	172.16.3.113	172.16.3.18	ICMP	98 Echo (ping) request id=0x1297, seq=6/1536, ttl=64 (reply in 29)
29.13.027651	172.16.3.18	172.16.3.113	ICMP	98 Echo (ping) reply id=0x1297, seq=6/1536, ttl=128 (request in 28)
30.14.051258	172.16.3.113	172.16.3.18	ICMP	98 Echo (ping) request id=0x1297, seq=7/1792, ttl=64 (reply in 31)
31.14.051454	172.16.3.18	172.16.3.113	ICMP	98 Echo (ping) reply id=0x1297, seq=7/1792, ttl=128 (request in 30)
32.14.644452	172.16.2.3	172.16.3.18	DNS	86 Standard query response 0xc0cd Server failure A slscr.update.microsoft.com
33.15.075391	172.16.3.113	172.16.3.18	ICMP	98 Echo (ping) request id=0x1297, seq=8/2048, ttl=64 (reply in 34)
34.15.075617	172.16.3.18	172.16.3.113	ICMP	98 Echo (ping) reply id=0x1297, seq=8/2048, ttl=128 (request in 33)
35.15.088536	172.16.3.18	172.16.2.3	DNS	86 Standard query 0x37b6 A slscr.update.microsoft.com
36.15.109305	172.16.3.18	172.16.2.3	DNS	86 Standard query 0x37b6 A slscr.update.microsoft.com
37.16.124910	172.16.3.18	172.16.2.3	DNS	86 Standard query 0x37b6 A slscr.update.microsoft.com
38.17.768477	172.16.3.113	172.16.3.18	TCP	74 37834 -> 80 [SYN] Seq=0 Win=29200 Len=0 MSS=1460 SACK_PERM=1 TSval=4138807592 TSecr=0 WS=128
39.18.140526	172.16.3.18	172.16.2.3	DNS	86 Standard query 0x37b6 A slscr.update.microsoft.com
40.18.187447	172.16.3.113	172.16.3.18	TCP	74 [TCP Retransmission] 37834 -> 80 [SYN] Seq=0 Win=29200 Len=0 MSS=1460 SACK_PERM=1 TSval=4138808611 TSecr=0 WS=128
41.19.194570	172.16.2.3	172.16.3.18	DNS	86 Standard query response 0x5e8f Server failure A slscr.update.microsoft.com
42.19.194571	172.16.2.3	172.16.3.18	DNS	86 Standard query response 0x5e8f Server failure A slscr.update.microsoft.com
43.19.194908	172.16.2.3	172.16.3.18	DNS	86 Standard query response 0x5e8f Server failure A slscr.update.microsoft.com
44.19.195502	172.16.2.3	172.16.3.18	DNS	86 Standard query response 0x5e8f Server failure A slscr.update.microsoft.com
45.19.765453	Intel_00:08:1b Intel_00:0c:01	ARP	42 Who has 172.16.3.1? Tell 172.16.3.18	
46.19.765826	Intel_00:0c:01 Intel_00:08:1b	ARP	60 172.16.3.1 is at 00:02:b3:00:0c:01	
47.20.003409	172.16.3.113	172.16.3.18	TCP	74 [TCP Retransmission] 37834 -> 80 [SYN] Seq=0 Win=29200 Len=0 MSS=1460 SACK_PERM=1 TSval=4138810627 TSecr=0 WS=128

Figure 7: WireShark from log

141	29.738407	172.16.3.113	172.16.3.18	TCP	60	48763	→	1007	[SYN]	Seq=0	Win=1024	Len=0	MSS=1460
142	29.738422	172.16.3.113	172.16.3.18	TCP	60	48763	→	1046	[SYN]	Seq=0	Win=1024	Len=0	MSS=1460
143	29.738422	172.16.3.113	172.16.3.18	TCP	60	48763	→	3905	[SYN]	Seq=0	Win=1024	Len=0	MSS=1460
144	29.738424	172.16.3.113	172.16.3.18	TCP	60	48763	→	1755	[SYN]	Seq=0	Win=1024	Len=0	MSS=1460
145	29.738426	172.16.3.113	172.16.3.18	TCP	60	48763	→	700	[SYN]	Seq=0	Win=1024	Len=0	MSS=1460
146	29.740637	172.16.3.113	172.16.3.18	TCP	60	48763	→	1121	[SYN]	Seq=0	Win=1024	Len=0	MSS=1460
147	29.740638	172.16.3.113	172.16.3.18	TCP	60	48763	→	1556	[SYN]	Seq=0	Win=1024	Len=0	MSS=1460
148	29.740639	172.16.3.113	172.16.3.18	TCP	60	48763	→	6001	[SYN]	Seq=0	Win=1024	Len=0	MSS=1460
149	29.740653	172.16.3.113	172.16.3.18	TCP	60	48763	→	32768	[SYN]	Seq=0	Win=1024	Len=0	MSS=1460
150	29.742778	172.16.3.113	172.16.3.18	TCP	60	48763	→	1328	[SYN]	Seq=0	Win=1024	Len=0	MSS=1460
151	29.742780	172.16.3.113	172.16.3.18	TCP	60	48763	→	7496	[SYN]	Seq=0	Win=1024	Len=0	MSS=1460
152	29.742796	172.16.3.113	172.16.3.18	TCP	60	48763	→	1580	[SYN]	Seq=0	Win=1024	Len=0	MSS=1460
153	29.742797	172.16.3.113	172.16.3.18	TCP	60	48763	→	1812	[SYN]	Seq=0	Win=1024	Len=0	MSS=1460
154	29.742798	172.16.3.113	172.16.3.18	TCP	60	48763	→	5925	[SYN]	Seq=0	Win=1024	Len=0	MSS=1460
155	29.742814	172.16.3.113	172.16.3.18	TCP	60	48763	→	19283	[SYN]	Seq=0	Win=1024	Len=0	MSS=1460
156	29.742835	172.16.3.113	172.16.3.18	TCP	60	48763	→	1147	[SYN]	Seq=0	Win=1024	Len=0	MSS=1460
157	29.742842	172.16.3.113	172.16.3.18	TCP	60	48763	→	8031	[SYN]	Seq=0	Win=1024	Len=0	MSS=1460
158	29.742889	172.16.3.113	172.16.3.18	TCP	60	48763	→	5120	[SYN]	Seq=0	Win=1024	Len=0	MSS=1460
159	29.742897	172.16.3.113	172.16.3.18	TCP	60	48763	→	3737	[SYN]	Seq=0	Win=1024	Len=0	MSS=1460
160	29.838540	172.16.3.113	172.16.3.18	TCP	60	48764	→	1007	[SYN]	Seq=0	Win=1024	Len=0	MSS=1460
161	29.838565	172.16.3.113	172.16.3.18	TCP	60	48764	→	1755	[SYN]	Seq=0	Win=1024	Len=0	MSS=1460
162	29.838566	172.16.3.113	172.16.3.18	TCP	60	48764	→	700	[SYN]	Seq=0	Win=1024	Len=0	MSS=1460
163	29.838567	172.16.3.113	172.16.3.18	TCP	60	48764	→	1046	[SYN]	Seq=0	Win=1024	Len=0	MSS=1460
164	29.838596	172.16.3.113	172.16.3.18	TCP	60	48764	→	4	[SYN]	Seq=0	Win=1024	Len=0	MSS=1460
165	29.838597	172.16.3.113	172.16.3.18	TCP	60	48764	→	10010	[SYN]	Seq=0	Win=1024	Len=0	MSS=1460
166	29.838609	172.16.3.113	172.16.3.18	TCP	60	48764	→	3905	[SYN]	Seq=0	Win=1024	Len=0	MSS=1460
167	29.840664	172.16.3.113	172.16.3.18	TCP	60	48764	→	1556	[SYN]	Seq=0	Win=1024	Len=0	MSS=1460
168	29.840715	172.16.3.113	172.16.3.18	TCP	60	48764	→	32768	[SYN]	Seq=0	Win=1024	Len=0	MSS=1460
169	29.840727	172.16.3.113	172.16.3.18	TCP	60	48764	→	1121	[SYN]	Seq=0	Win=1024	Len=0	MSS=1460
170	29.840729	172.16.3.113	172.16.3.18	TCP	60	48764	→	6001	[SYN]	Seq=0	Win=1024	Len=0	MSS=1460
171	29.842782	172.16.3.113	172.16.3.18	TCP	60	48764	→	19283	[SYN]	Seq=0	Win=1024	Len=0	MSS=1460
172	29.842886	172.16.3.113	172.16.3.18	TCP	60	48764	→	1812	[SYN]	Seq=0	Win=1024	Len=0	MSS=1460
173	29.842888	172.16.3.113	172.16.3.18	TCP	60	48764	→	1580	[SYN]	Seq=0	Win=1024	Len=0	MSS=1460
174	29.842891	172.16.3.113	172.16.3.18	TCP	60	48764	→	1147	[SYN]	Seq=0	Win=1024	Len=0	MSS=1460
175	29.842892	172.16.3.113	172.16.3.18	TCP	60	48764	→	1328	[SYN]	Seq=0	Win=1024	Len=0	MSS=1460
176	29.842893	172.16.3.113	172.16.3.18	TCP	60	48764	→	7496	[SYN]	Seq=0	Win=1024	Len=0	MSS=1460
177	29.844969	172.16.3.113	172.16.3.18	TCP	60	48764	→	5925	[SYN]	Seq=0	Win=1024	Len=0	MSS=1460
178	29.844970	172.16.3.113	172.16.3.18	TCP	60	48764	→	8031	[SYN]	Seq=0	Win=1024	Len=0	MSS=1460
179	29.844983	172.16.3.113	172.16.3.18	TCP	60	48764	→	5120	[SYN]	Seq=0	Win=1024	Len=0	MSS=1460
180	29.844985	172.16.3.113	172.16.3.18	TCP	60	48764	→	3737	[SYN]	Seq=0	Win=1024	Len=0	MSS=1460
181	29.938684	172.16.3.113	172.16.3.18	TCP	60	48763	→	14441	[SYN]	Seq=0	Win=1024	Len=0	MSS=1460
182	29.938690	172.16.3.113	172.16.3.18	TCP	60	48763	→	8085	[SYN]	Seq=0	Win=1024	Len=0	MSS=1460
183	29.938690	172.16.3.113	172.16.3.18	TCP	60	48763	→	32771	[SYN]	Seq=0	Win=1024	Len=0	MSS=1460
184	29.938691	172.16.3.113	172.16.3.18	TCP	60	48763	→	65000	[SYN]	Seq=0	Win=1024	Len=0	MSS=1460
185	29.938691	172.16.3.113	172.16.3.18	TCP	60	48763	→	7999	[SYN]	Seq=0	Win=1024	Len=0	MSS=1460
186	29.938736	172.16.3.113	172.16.3.18	TCP	60	48763	→	2968	[SYN]	Seq=0	Win=1024	Len=0	MSS=1460
187	29.938736	172.16.3.113	172.16.3.18	TCP	60	48763	→	6839	[SYN]	Seq=0	Win=1024	Len=0	MSS=1460

Figure 8: Half-open scan

```

AxCutemRule1 - Notepad
File Edit Format View Help
alert tcp $HOME_NET any -> |$HOME_NET 445 (msg:"Possible CVE-2025-24054 NTLM hash leak via SMB"; flow:established,to_server; content:"NTLMSSP"; offset:0; depth:100; sid:10000001; rev:1;)

```

Figure 9: The first custom rule



Figure 10: My second custom rule

```

[ Port Based Pattern Matching Memory ]
-- [ Aho-Corasick Summary ] -----
Storage Format      : Full-Q
Finite Automaton    : DFA
Alphabet Size       : 256 Chars
Sizeof State        : Variable (1,2,4 bytes)
Instances           : 119
  1 byte states     : 108
  2 byte states     : 11
  4 byte states     : 0
Characters          : 166377
States              : 132522
Transitions         : 24183483
State Density       : 71.3%
Patterns            : 7958
Match States        : 8429
Memory (MB)         : 66.97
  Patterns          : 0.68
  Match Lists       : 1.13
  DFA
    1 byte states   : 0.69
    2 byte states   : 64.35
    4 byte states   : 0.00
-----
[ Number of patterns truncated to 20 bytes: 454 ]
pcap DAQ configured to passive.
The DAQ version does not support reload.
Acquiring network traffic from "\Device\NPF_{B09EB7D3-148C-446C-B745-B2B082CED37A}".
Decoding Ethernet

--== Initialization Complete ==--

o+~)~
....

-*> Snort! <*-
Version 2.9.17-WIN32 GRE (Build 199)
By Martin Roesch & The Snort Team: http://www.snort.org/contact#team
Copyright (C) 2014-2020 Cisco and/or its affiliates. All rights reserved.
Copyright (C) 1998-2013 Sourcefire, Inc., et al.
Using PCRE version: 8.10 2010-06-25
Using ZLIB version: 1.2.3

Rules Engine: SF_SNORT_DETECTION_ENGINE Version 3.1 <Build 1>
Preprocessor Object: SF_SSLPP Version 1.1 <Build 4>
Preprocessor Object: SF_SSH Version 1.1 <Build 3>
Preprocessor Object: SF_SMTP Version 1.1 <Build 9>
Preprocessor Object: SF_SIP Version 1.1 <Build 1>
Preprocessor Object: SF_SOF Version 1.1 <Build 1>
Preprocessor Object: SF_REPUTATION Version 1.1 <Build 1>
Preprocessor Object: SF_POP Version 1.0 <Build 1>
Preprocessor Object: SF_MODBUS Version 1.1 <Build 1>
Preprocessor Object: SF_IMAP Version 1.0 <Build 1>
Preprocessor Object: SF_GTP Version 1.1 <Build 1>
Preprocessor Object: SF_FTPTELNET Version 1.2 <Build 13>
Preprocessor Object: SF_DNS Version 1.1 <Build 4>
Preprocessor Object: SF_DNP3 Version 1.1 <Build 1>
Preprocessor Object: SF_DCERPC2 Version 1.0 <Build 3>
Commencing packet processing (pid=6476)

```

Figure 11: SNORT running with both rules active